

Auteur: Stan Van Campenhout

Studierichting: Informatica

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$$\begin{aligned} \int \sqrt{1 - \sqrt{x}} dx &\stackrel{t=\sqrt{x}}{=} \int \sqrt{1 - t} \cdot 2t dt = 2 \int t \sqrt{1 - t} dt \\ &\stackrel{u=\sqrt{1-t}}{=} 2 \int (1 - u^2) \cdot u \cdot (-2u) du = -4 \int (u^2 - u^4) du \\ &= -\frac{4}{3}u^3 + \frac{4}{5}u^5 + c \stackrel{t=\sqrt{1-\sqrt{x}}}{=} -\frac{4}{3}\sqrt{(1 - \sqrt{x})^3} + \frac{4}{5}\sqrt{(1 - \sqrt{x})^5} + c \end{aligned}$$

References